Weather Market Edge

A calibrated probability engine for Kalshi high-temperature contracts.

Edge sources	**3**
Settlement stations, v1	**6**
Forecast horizon	**T+36h**
Data cost	**$0 — public feeds**
Build to first signal	**4 weeks**

NWS API · Iowa Environmental Mesonet · NBM / HRRR via NOAA AWS · Kalshi public API

THE THESIS

The market prices a phone-app forecast. We price a distribution.

01

Station-specific bias

Central Park does not behave like an airport. KNYC's urban siting and 1-minute sensor produce a repeatable offset against the NWS point forecast — measurable, seasonal, and free to compute.

+0.8°F mean summer bias, KNYC

02

Probabilistic model output

NBM ships a full percentile spread for daily max temperature. A bracket market is a question about spread, not about the median — and the median is the only number most participants see.

P10–P90 width drives every bracket

03

Intraday lock-in

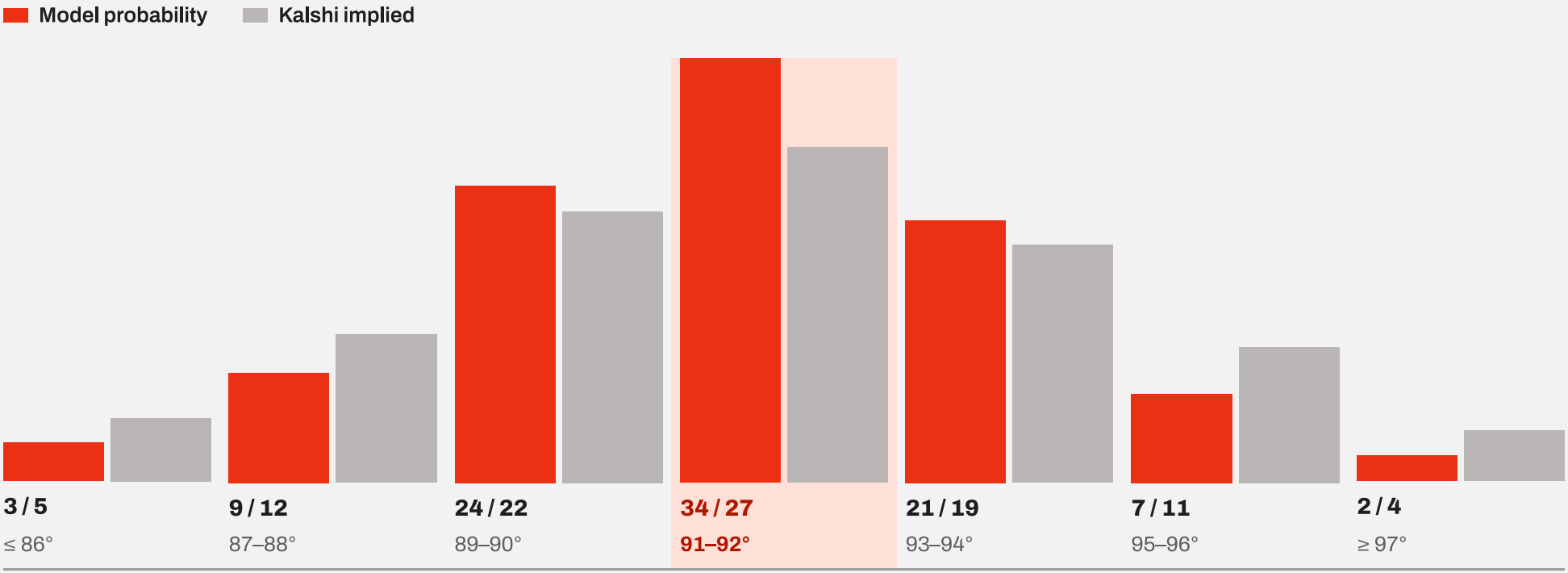
Once a running high is touched, brackets below it are settled facts. The market often keeps quoting them at 3–8¢ for hours. That is arithmetic, not forecasting.

3 of 7 brackets dead by 14:00 LST

EDGE SOURCE 01 + 02

Calibrated bracket distribution

KNYC · 24 Jul 2026
Bias-corrected NBM

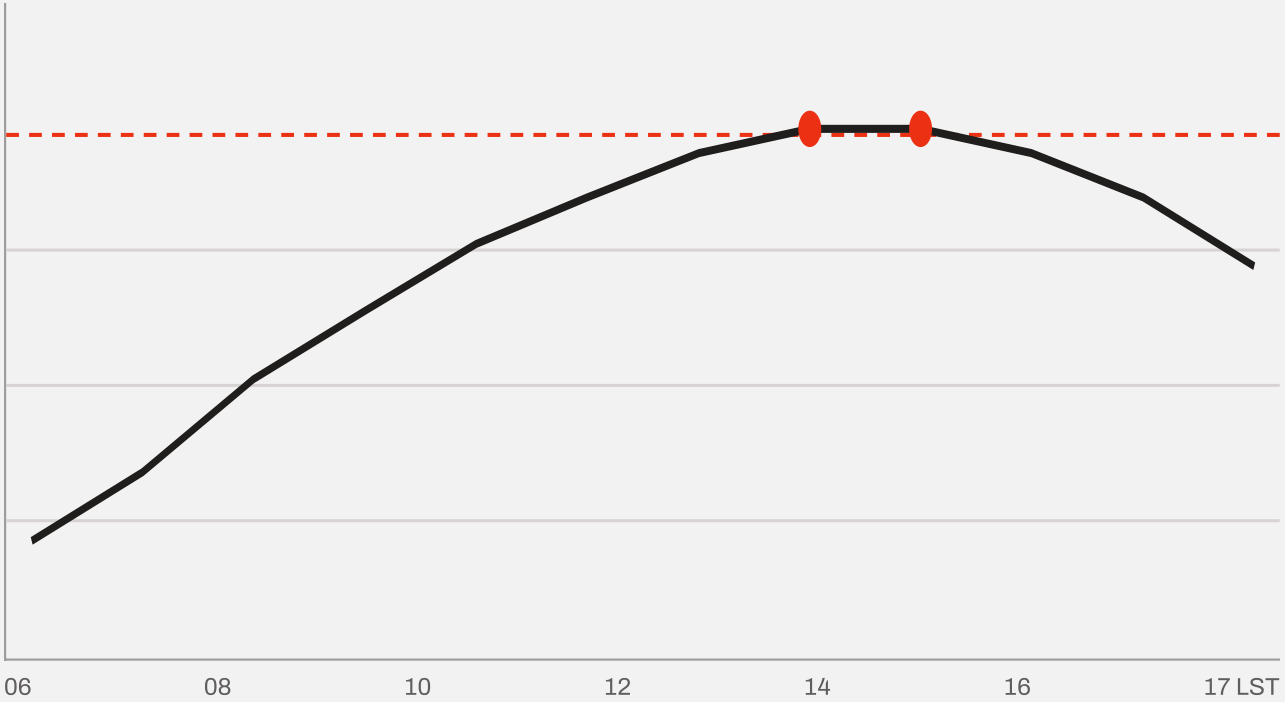


NBM MEDIAN	BIAS APPLIED	Σ (CALIBRATED)	BEST DIVERGENCE
91.2°F	**+0.8°F**	**1.9°F**	**+7.0 pts**

EDGE SOURCE 03

Running high & mathematical lock-in

Hourly ASOS, KNYC
Running high 92°F at 14:00 LST



BRACKET STATUS, 14:07 LST	
~~≤ 86°~~	DEAD
~~87–88°~~	DEAD
~~89–90°~~	DEAD
91–92°	LIVE 61%
93–94°	LIVE 31%
95–96°	LIVE 7%
≥ 97°	LIVE 1%

Caveat surfaced in the UI: the running high is read from rounded METAR, while settlement uses the 1-minute sensor via CLI. A 91.6°F peak reports as 92 in METAR and can settle either side — we mark those cases **ambiguous** rather than locked.

Still quoted at 3–6¢ on the exchange: three settled brackets.

MODEL MINUS MARKET

Where we disagree with the book

Actionable threshold
±5 points

STATION	BRACKET	MODEL	MARKET	DIVERGENCE	SIGNAL
KAUS	101–102°	41%	33%	+8.0	ACTIONABLE
KMDW	88–89°	29%	21%	+8.0	ACTIONABLE
KNYC	91–92°	34%	27%	+7.0	ACTIONABLE
KDFW	99–100°	18%	25%	−7.0	ACTIONABLE
KLAX	79–80°	12%	19%	−7.0	ACTIONABLE
KPHL	93–94°	26%	24%	+2.0	below threshold

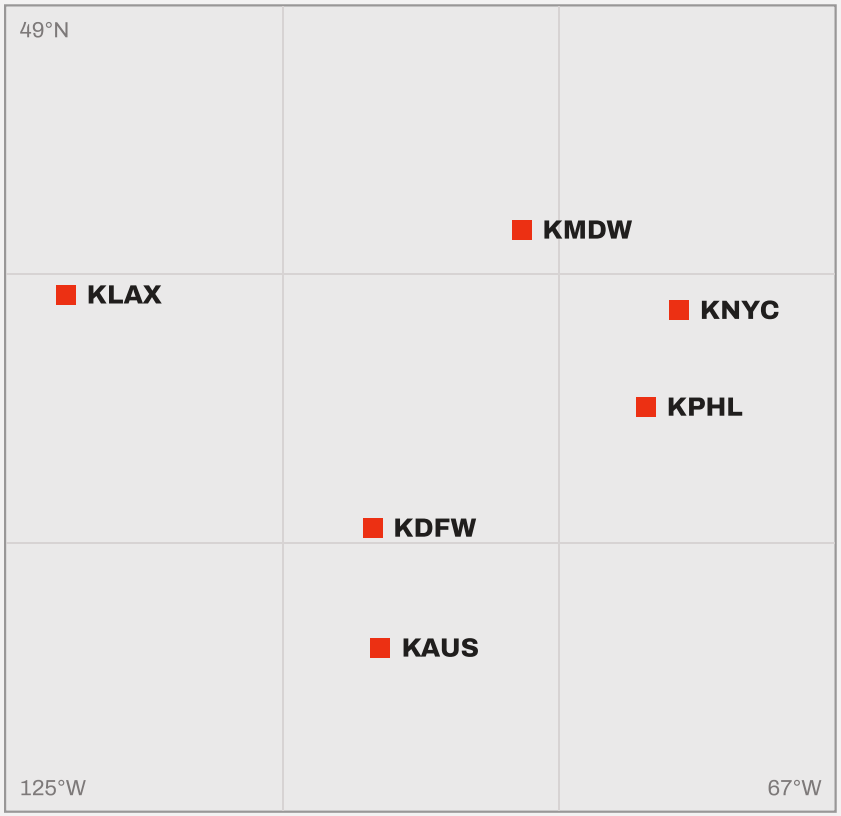
Divergence is reported in probability points, not cents — fee and spread drag are applied separately in the trade view. Anything inside the threshold is shown but never flagged.

COVERAGE & CALIBRATION

Six settlement stations

90-day rolling window
IEM / ACIS forecast–CLI pairs

RELATIVE POSITION — LON / LAT GRID



ID	SITE	BIAS	RMSE	TZ (LST)
KNYC	Central Park, NY	**+0.8**	1.9	UTC–5
KMDW	Chicago Midway, IL	**−0.3**	2.4	UTC–6
KAUS	Austin–Bergstrom, TX	**+1.4**	2.1	UTC–6
KDFW	Dallas–Fort Worth, TX	**+0.6**	2.6	UTC–6
KLAX	Los Angeles Intl, CA	**−1.1**	1.7	UTC–8
KPHL	Philadelphia Intl, PA	**+0.2**	2.2	UTC–5

Bias in °F against the NWS point forecast, summer season. A station only enters the actionable set once it has ≥60 valid pairs in the window.

PART TWO

The product

Dashboard Market board Contract view Architecture

SCREEN 01 — STATION DASHBOARD

1440 × 900 · desktop

WX EDGE | Dashboard | Markets | Calibration

Obs 14:07 LST · 8 min ago | Refresh

STATIONS

KNYC	+7.0
KAUS	+8.0
KMDW	+8.0
KDFW	−7.0
KLAX	−7.0
KPHL	+2.0

Sorted by absolute divergence. Grey rows sit inside threshold.

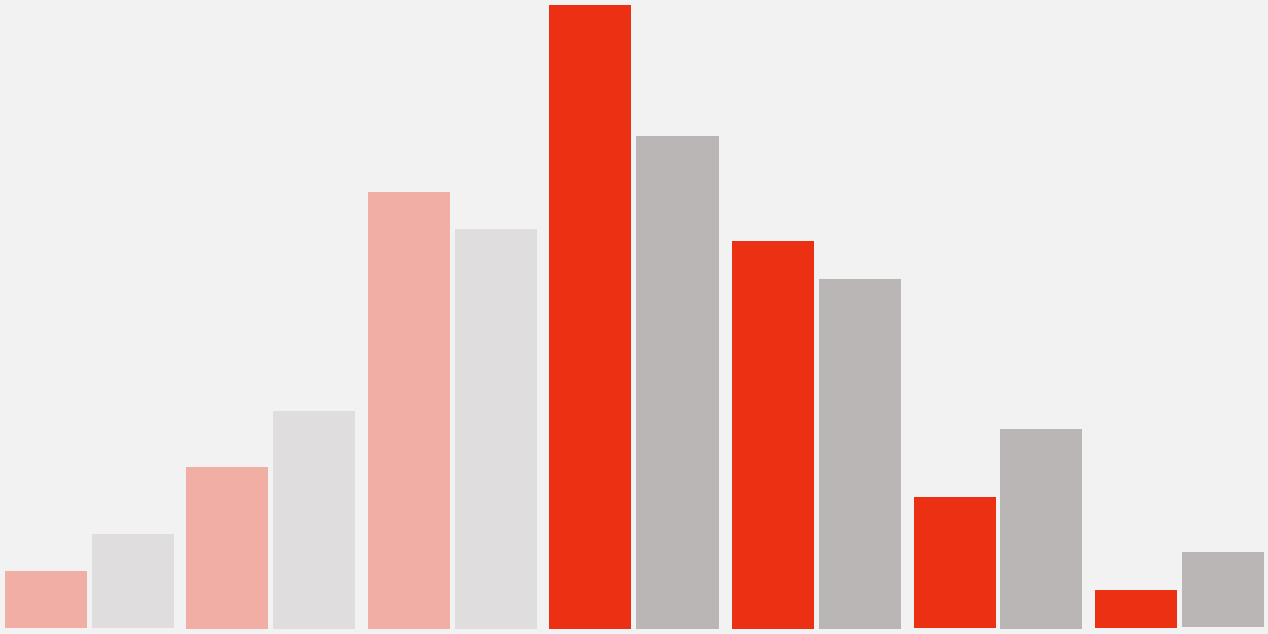
CENTRAL PARK, NEW YORK

KNYC · 24 Jul

RUNNING HIGH

92°F

BRACKET PROBABILITY — MODEL VS IMPLIED



Faded = mathematically eliminated Model 34% · Market 27%

SETTLEMENT WINDOW

00:00 – 23:59 LST

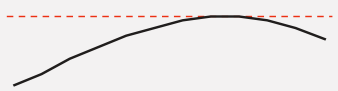
= 01:00 – 00:59 local clock

DST active — window spills into 25 Jul

TIME TO SETTLE

09h 52m

INTRADAY



CALIBRATION

Mean bias	**+0.8°F**
RMSE	**1.9°F**
Pairs (90d)	**88**

Open KNYC market board

SCREEN 02 — MARKET BOARD

Ranked by fee-adjusted edge · view only

WX EDGE Dashboard **Markets** Calibration

All | Actionable | Locked

MARKET	BRACKET	MODEL	BID / ASK	IMPLIED	EDGE (NET)	STATUS
KXHIGHAUS	101–102°	**41%**	32 / 34	33%	**+6.4**	**ACTIONABLE**
KXHIGHCHI	88–89°	**29%**	20 / 23	21%	**+5.9**	**ACTIONABLE**
KXHIGHNY	91–92°	**34%**	26 / 28	27%	**+5.1**	**ACTIONABLE**
KXHIGHNY	~~89–90°~~	**0%**	3 / 6	4%	**−4.0**	**SETTLED · SHORT**
KXHIGHDFW	99–100°	**18%**	24 / 27	25%	**−5.6**	**ACTIONABLE**
KXHIGHLAX	79–80°	**12%**	18 / 21	19%	**−5.2**	**ACTIONABLE**
KXHIGHPHIL	93–94°	**26%**	23 / 26	24%	**+0.3**	**BELOW THRESHOLD**
KXHIGHAUS	99–100°	**22%**	21 / 24	22%	**−1.1**	**BELOW THRESHOLD**

Net edge = model probability − ask, less 1.0¢ round-trip fee assumption.

Book snapshot 14:07:12 LST · 22s latency

SCREEN 03 — CONTRACT DETAIL

Analysis only — no order placement in v1

KXHIGHNY · 24 JUL 2026

High temp 91–92°F

Model **34%** Ask **28¢** Net edge **+5.1 pts**

WHY THE MODEL DISAGREES

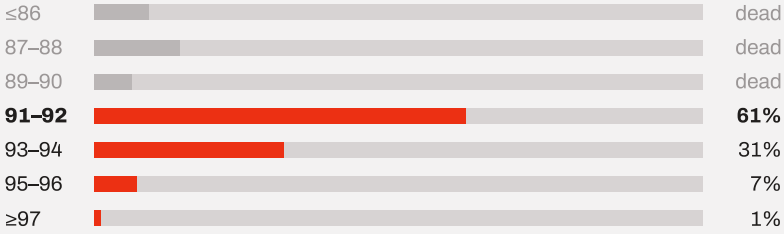
Raw NBM median	**90.4°F**
KNYC summer bias correction	**+0.8°F**
Calibrated median	**91.2°F**
Bracket mass 91.0–92.9°F	**34%**
Floor from running high (92°F touched)	**≥ 61% conditional**

The market's 28¢ is consistent with an uncorrected point forecast of 90°F. Applying the station bias moves mass one bracket up; the touched running high removes the downside tail entirely.

ORDER BOOK

PRICE	SIZE	SIDE
31¢	180	ask
30¢	420	ask
28¢	**1,150**	**best ask**
26¢	**960**	**best bid**
25¢	640	bid
23¢	310	bid

MODEL VS IMPLIED, ALL BRACKETS



Copy analysis | Open on Kalshi

ARCHITECTURE

Four public feeds, one distribution

INGEST

NWS API
Hourly obs + point forecast per station

NBM / HRRR
Percentile temperature spread, NOAA AWS

IEM / ACIS
90-day forecast–CLI pairs, ASOS archive

Kalshi public API
Order book → implied probability

COMPUTE

Calibration module
Mean bias + RMSE per station per season

Distribution builder
Bias-shifted percentiles → continuous CDF

Bracket mapper
CDF integrated over Kalshi's exact edges

Lock-in resolver
Running high → eliminated / guaranteed set

SERVE

OpenAPI contract
Stations · detail · distribution · prices · bias

Codegen client
Orval → typed hooks, no hand-written fetch

Cache layer
Obs 5 min · NBM 1 h · bias nightly

SURFACE

React + Vite
Three screens, no auth, no persistence

Polling
Book 30 s · obs 5 min, visible timestamps

FAILURE MODE

Any stale feed suppresses the signal rather than degrading it silently.

THE TRAP

Settlement runs on standard time, not your clock.

CLI daily maximum is computed over the local **standard**-time day. Between March and November that window is offset one hour from the wall clock — so an evening spike after 23:00 local can land in the next contract's day, and the last hour of the settlement day has already happened before midnight.

The dashboard prints both windows, side by side, always. No inference required from the user.

KNYC · CONTRACT 24 JUL 2026

Settlement window (LST)	**00:00 → 23:59**
Same window, local clock	**01:00 → 00:59^{+1d}**
UTC	**05:00 → 04:59**
Hours remaining	**09h 52m**

WHERE THE HOUR GOES



Local-clock day, 24 Jul → spills to 25 Jul

Every timestamp in the app carries an explicit **LST** or **local** suffix. There is no bare time anywhere in the interface.

One number, two calendars — surfaced, not explained away.

BUILD PLAN

Four sprints to first live signal

One engineer
Weeks of 27 Jul → 17 Aug

WEEK 1 — 27 Jul

Contract & skeleton

- React + Vite app at root
- openapi.yaml: 5 endpoints
- Orval codegen wired
- NWS obs + point forecast live

Exit: real observations on screen

WEEK 2 — 3 Aug

Distribution engine

- NBM percentile ingest from AWS
- IEM 90-day pair backfill
- Bias + RMSE per station
- CDF → bracket integration

Risk: GRIB parsing cost

WEEK 3 — 10 Aug

Market & lock-in

- Kalshi book → implied probs
- Bracket-ID matching per station
- Running-high lock-in resolver
- METAR-vs-CLI ambiguity flag

Exit: first signed edge number

WEEK 4 — 17 Aug

Dashboard & timing

- Three screens built out
- LST / local dual clock
- Staleness suppression
- Paper-track log for scoring

Exit: 2 weeks of paper results

BOUNDARIES

What v1 is not, and what could go wrong

OUT OF SCOPE

- Automated trading or order placement
- Non-US and non-Kalshi stations
- Precipitation and other weather markets
- Calibration beyond T+36h
- Accounts, portfolios, saved state

Each of these is a deliberate cut, not an omission — the point of v1 is to establish whether the edge survives contact with real prices.

RISKS

GRIB decode cost — HIGH

NBM files are large; per-point extraction may need a cached subset job.

Bracket-ID drift — HIGH

Kalshi re-cuts bracket edges seasonally; mapping must be derived, never hard-coded.

Thin books — MED

A 7-point edge on 40 contracts of depth is not a real opportunity; size must be shown next to edge.

Overfit bias window — MED

90 days of summer pairs is thin; a regime change reads as bias.

Feed outage — LOW

Handled: staleness suppresses the signal instead of showing an old one.

THE ASK

Four weeks, one engineer, then a paper-results gate.

Go / no-go	**31 Aug 2026**
Pass bar	**Brier < market, 2 weeks**
Kill condition	**No bracket > 5 pts edge**
Next decision	**Sizing & execution scope**

View-only edge analysis · No order placement in v1 · All figures illustrative